GA-01 · Physics (NCERT Class 9–12 Level)

RRB JE CBT-2 · Electronics & Allied Engineering Group (CEN 05/2025)

Section: Physics & Chemistry — 15 Questions Combined

> **How to use this chapter.** The CBT-2 "Physics & Chemistry" section sits at NCERT Class 10–12 level, with a strong tilt toward Class 10 concepts plus a few Class 11–12 fundamentals. Every fact in this chapter is aligned with NCERT Science (Class 9 & 10) and NCERT Physics (Class 11 & 12). Work through the theory, the 12 worked examples, the PYQ Corner (real 2025 CBT-2 questions), and finish with the 25 MCQs and the formula sheet.

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1. Units, Measurement & Errors

1.1 The SI System (NCERT Class 11, Ch. Units and Measurements)

The SI system has **seven base units**:

Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Two supplementary units: radian (plane angle) and steradian (solid angle).

1.2 Common Derived Units

Quantity	Unit	In base units
Force	newton (N)	$kg \cdot m \cdot s^{-2}$
Work / Energy	joule (J)	$kg \cdot m^2 \cdot s^{-2}$
Power	watt (W)	$kg \cdot m^2 \cdot s^{-3}$
Pressure	pascal (Pa)	$kg \cdot m^{-1} \cdot s^{-2}$
Frequency	hertz (Hz)	s^{-1}
Charge	coulomb (C)	$A \cdot s$
Potential difference	volt (V)	$J \cdot C^{-1}$
Resistance	ohm (Ω)	$V \cdot A^{-1}$
Power of lens	dioptre (D)	m^{-1}

1.3 Errors in Measurement

- **Absolute error** = |true value − measured value|.
- **Relative error** = mean absolute error ÷ mean value.
- **Percentage error** = relative error × 100.
- In **addition/subtraction**, absolute errors add: $\Delta z = \Delta a + \Delta b$.
- In **multiplication/division**, relative errors add: $\Delta z/z = \Delta a/a + \Delta b/b$.
- For a power, $z = a^n \Rightarrow \Delta z/z = n(\Delta a/a)$.

Significant figures: all non-zero digits are significant; zeros between non-zero digits are significant; trailing zeros after a decimal point are significant (2.500 has 4 sig. figs.).

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2. Motion

2.1 Basic Definitions (NCERT Class 9, Ch. Motion)

Quantity	Definition	Nature
Distance	Actual path length covered	Scalar
Displacement	Shortest directed line from initial to final position	Vector
Speed	Distance ÷ time	Scalar
Velocity	Displacement ÷ time	Vector
Acceleration	Change in velocity ÷ time	Vector

Displacement can be zero even when distance is not (e.g., one complete round of a circular track).

2.2 Equations of Uniformly Accelerated Motion

For initial velocity u, final velocity v, acceleration a, time t, displacement s:

1. **$v = u + at$**
2. **$s = ut + \frac{1}{2}at^2$**
3. **$v^2 = u^2 + 2as$**

Distance in the nth second: **$s_n = u + a(2n - 1)/2$**

2.3 Motion Graphs

Graph	Slope gives	Area under graph gives
Distance–time	Speed	—
Velocity–time	Acceleration	Displacement
Acceleration–time	—	Change in velocity

- Straight horizontal line on v–t graph → uniform velocity (zero acceleration).
- Straight sloping line on v–t graph → uniform acceleration.
- Parabolic s–t graph → uniformly accelerated motion.

2.4 Uniform Circular Motion

Motion at constant speed on a circle is **accelerated** because direction (hence velocity) changes continuously. Centripetal acceleration $a = v^2/r$, directed toward the centre.

2.5 Projectile Basics (Class 11)

For a projectile launched with speed u at angle θ (air resistance neglected):

- Time of flight: $T = 2u \sin\theta / g$
- Maximum height: $H = u^2 \sin^2\theta / 2g$
- Horizontal range: $R = u^2 \sin 2\theta / g$ — **maximum at θ = 45°**
- Ranges are equal for complementary angles (θ and 90° − θ).
- The trajectory is a **parabola**; horizontal velocity remains constant.

Worked Example 1. A train starts from rest and attains 72 km/h in 40 s. Find its acceleration and the distance covered.

u = 0, v = 72 km/h = 20 m/s, t = 40 s.

a = (v − u)/t = 20/40 = **0.5 m/s²**.

s = ut + ½at² = 0 + ½ × 0.5 × 1600 = **400 m**.

Worked Example 2. A stone is dropped from a 45 m tower (g = 10 m/s²). Find the time to reach the ground and the striking velocity.

s = ½gt² ⇒ 45 = ½ × 10 × t² ⇒ t² = 9 ⇒ **t = 3 s**.

v = gt = 10 × 3 = **30 m/s**.

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3. Laws of Motion & Friction

3.1 Newton's Laws (NCERT Class 9, Ch. Force and Laws of Motion)

- **First law (law of inertia):** A body remains at rest or in uniform motion in a straight line unless acted upon by an external unbalanced force. Inertia is measured by **mass**.
- **Second law:** Rate of change of momentum ∝ applied force; **F = ma**. Momentum **p = mv** (SI unit kg·m/s).
- **Third law:** To every action there is an equal and opposite reaction, acting on **different bodies** simultaneously.

Conservation of momentum: In the absence of external force, total momentum of a system is constant: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ (basis of rocket propulsion and gun recoil).

Impulse = F × Δt = change in momentum (unit N·s). Catching a ball by pulling hands back increases Δt, reducing force.

3.2 Friction (NCERT Class 8/11)

- Friction opposes **relative motion** between surfaces in contact; it acts tangentially.
- Order of magnitude: **static > sliding (kinetic) > rolling** friction. Ball bearings exploit low rolling friction.
- Limiting static friction f = μN, where μ = coefficient of friction, N = normal reaction. It is **independent of apparent contact area** (for given surfaces and normal force).
- Friction is a **necessary evil**: it enables walking, braking, writing, but wastes energy as heat and causes wear. Lubricants and streamlining reduce it.

Worked Example 3. A force of 20 N acts on a 4 kg body at rest for 5 s. Find the final velocity.

a = F/m = 20/4 = 5 m/s². v = u + at = 0 + 5 × 5 = **25 m/s**.

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4. Work, Power & Energy

4.1 Work

W = F s cos θ (θ = angle between force and displacement). SI unit: joule (1 J = 1 N·m).

- θ = 0° → W = Fs (maximum positive work).
- θ = 90° → W = 0 (e.g., gravity on a horizontally moving body; centripetal force in circular motion does **no work**).
- θ = 180° → negative work (friction on a moving body).

4.2 Kinetic Energy — Derivation from $v^2 = u^2 + 2as$

Consider a body of mass m accelerated from rest (u = 0) to velocity v by force F over displacement s:

$v^2 = u^2 + 2as \Rightarrow v^2 = 2as \Rightarrow s = v^2/2a$

Work done $W = Fs = (ma)(v^2/2a) = \mathbf{\frac{1}{2}mv^2}$

This work is stored as kinetic energy: **$KE = \frac{1}{2}mv^2$**. Also, **$KE = p^2/2m$** (p = momentum).

4.3 Potential Energy

Work done in raising mass m through height h against gravity: **PE = mgh**.

4.4 Conservation of Mechanical Energy

For a freely falling body (no air resistance), **KE + PE = constant** at every point. At the top: all PE; at the bottom: all KE; total mgh throughout.

Work–energy theorem: Net work done on a body = change in its kinetic energy.

4.5 Power

P = W/t = F·v. SI unit: watt (1 W = 1 J/s). 1 horsepower (hp) = 746 W.

Commercial unit of energy: 1 kilowatt-hour (kWh) = 1 "unit" = 3.6×10^6 J.

> **Worked Example 4.** A 2 kg body moving at 10 m/s is brought to rest in 4 m. Find the retarding force.
>
> Using $v^2 = u^2 + 2as$: $0 = 100 + 2a(4) \Rightarrow a = -12.5$ m/s².
>
> F = ma = 2 × 12.5 = **25 N** (opposing motion).
>
> (Check by work–energy theorem: KE lost = ½ × 2 × 10^2 = 100 J = F × 4 ⇒ F = 25 N. ✓)

> **Worked Example 5.** A pump lifts 200 kg of water through 10 m in 20 s (g = 10 m/s²). Find its power.
>
> W = mgh = 200 × 10 × 10 = 20,000 J. P = W/t = 20,000/20 = **1000 W = 1 kW**.

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5. Gravitation

5.1 Universal Law (NCERT Class 9, Ch. Gravitation)

$F = G\,m_1m_2 / r^2$, with G = 6.674×10^{-11} N·m²/kg² (universal gravitational constant, measured by Cavendish).

5.2 g vs G

	g	G
Name	Acceleration due to gravity	Universal gravitational constant
Value	≈ 9.8 m/s^2 on Earth's surface	6.674×10^{-11} $N{\cdot}m^2/kg^2$
Nature	Varies with location (altitude, depth, latitude)	Universal constant, same everywhere
Relation	$g = GM/R^2$	—
Unit	m/s^2	$N{\cdot}m^2/kg^2$

- g is **maximum at the poles, minimum at the equator**; decreases with altitude and with depth; **zero at Earth's centre**.
- Mass is constant everywhere; **weight W = mg** varies with g. Weight on the Moon ≈ 1/6 of that on Earth.

5.3 Kepler's Laws of Planetary Motion

1. **Law of orbits:** Every planet moves in an **ellipse** with the Sun at one focus.
2. **Law of areas:** The line joining planet and Sun sweeps **equal areas in equal times** (consequence of conservation of angular momentum) — planets move faster when nearer the Sun.
3. **Law of periods: $T^2 \propto r^3$** (square of orbital period ∝ cube of semi-major axis).

Escape velocity: $v_e = \sqrt{(2gR)} \approx$ **11.2 km/s** for Earth. Orbital velocity near Earth's surface ≈ 7.9 km/s; $v_e = \sqrt{2} \times v_o$.

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6. Pressure & Floatation

6.1 Pressure

P = F/A (thrust per unit area). SI unit: pascal (Pa). Sharp knives and pointed nails work by reducing area to increase pressure.

Pressure at depth h in a liquid of density ρ: **P = hρg** (same in all directions at a point — liquids transmit pressure).

Atmospheric pressure ≈ 1.013×10^5 Pa = 760 mm of Hg (measured with a barometer).

6.2 Pascal's Law

Pressure applied to an enclosed fluid is transmitted **undiminished** to every part of the fluid and the walls of the container. Applications: **hydraulic press, hydraulic brakes, hydraulic lift** — a small force on a small piston produces a large force on a large piston ($F_2 = F_1 \times A_2/A_1$).

6.3 Archimedes' Principle & Floatation

When a body is immersed (fully or partially) in a fluid, it experiences an **upthrust (buoyant force) equal to the weight of the fluid displaced**.

- Body floats if its density ≤ density of the fluid; sinks if greater.
- Floating body: weight of body = weight of liquid displaced (law of floatation).
- Applications: ships, submarines, hydrometers (measure liquid density), lactometers (test milk purity).
- **Relative density** = density of substance ÷ density of water (no unit). RD of iron ≈ 7.8; ice floats because RD of ice ≈ 0.92.

> **Worked Example 6.** A solid weighs 60 N in air and 40 N fully immersed in water. Find the buoyant force and the relative density of the solid.

> Buoyant force = 60 − 40 = **20 N** = weight of water displaced.

> Relative density = weight in air ÷ loss of weight in water = 60/20 = **3**.

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7. Elasticity & Hooke's Law

- **Elasticity:** property of regaining original shape/size after the deforming force is removed. Quartz is nearly perfectly elastic; putty is nearly perfectly plastic. **Steel is more elastic than rubber** (steel needs more stress for the same strain).
- **Stress** = restoring force per unit area (N/m^2); **strain** = fractional change (no unit).
- **Hooke's law:** within the elastic limit, **stress ∝ strain**; stress/strain = modulus of elasticity.
- **Young's modulus Y** = longitudinal stress/longitudinal strain = (F/A)/(ΔL/L).
- **Bulk modulus** (volume), **Shear modulus** (shape).
- For a spring: **F = kx** (k = spring constant); elastic PE stored = ½kx^2.
- Beyond the **elastic limit**, permanent deformation occurs; at the **breaking point**, the material fractures.

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8. Heat & Thermal Physics

8.1 Temperature Scales

C/5 = (F − 32)/9 = (K − 273.15)/5. The Celsius and Fahrenheit scales agree at **−40°**. SI unit of temperature: kelvin.

8.2 Modes of Heat Transmission

Mode	Medium needed?	Mechanism	Example
Conduction	Yes (mainly solids)	Particle-to-particle transfer, no bulk movement	Metal spoon in hot tea
Convection	Yes (liquids/gases)	Bulk movement of heated fluid	Sea breeze, room heater
Radiation	No (works in vacuum)	Electromagnetic waves (infrared)	Heat of the Sun

Black rough surfaces are good absorbers **and** good emitters; shiny surfaces are poor absorbers/good reflectors (thermos flask uses silvered walls + vacuum to cut all three modes).

8.3 Specific Heat & Latent Heat

- **Specific heat capacity c:** heat needed to raise 1 kg by 1 K. **Q = mcΔT.** Water has an unusually high c = 4186 J/(kg·K) ≈ 1 cal/(g·°C) — hence its use as a coolant and the moderating effect of seas on climate.
- **Latent heat L:** heat absorbed/released during change of state at constant temperature. **Q = mL.**
- Latent heat of fusion of ice ≈ 3.34 × 10^5 J/kg (80 cal/g).
- Latent heat of vaporisation of water ≈ 22.6 × 10^5 J/kg (540 cal/g).
- Steam at 100 °C causes worse burns than water at 100 °C because it carries the extra latent heat of vaporisation.
- **Evaporation** is a surface phenomenon occurring at all temperatures; it causes cooling (sweating, earthen pots).

8.4 Thermal Expansion

Linear expansion: ΔL = LαΔT; area (β ≈ 2α); volume (γ ≈ 3α). Applications: gaps in railway tracks, sagging telephone wires in summer, bimetallic strips in thermostats.

8.5 Anomalous Expansion of Water

Water **contracts** on heating from 0 °C to 4 °C and expands beyond 4 °C. Hence water has **maximum density at 4 °C**. In winter, ponds freeze from the **top**, while water below stays at 4 °C — aquatic life survives under the insulating ice layer.

> **Worked Example 7.** How much heat is needed to convert 2 kg of ice at 0 °C completely into water at 0 °C? ($L_{fusion} = 3.34 \times 10^5$ J/kg)

> $Q = mL = 2 \times 3.34 \times 10^5 =$ **6.68×10^5 J**. Temperature does not change during melting.

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9. Sound

9.1 Nature & Propagation (NCERT Class 9, Ch. Sound)

- Sound is a **mechanical, longitudinal** wave (compressions and rarefactions) — it **cannot travel through vacuum**.
- Speed order: **$v_{solid} > v_{liquid} > v_{gas}$**. In air at 20–25 °C, v ≈ 340–346 m/s; in water ≈ 1480 m/s; in steel ≈ 5000 m/s. Speed in air increases with temperature and humidity.
- **$v = f\lambda$** (speed = frequency × wavelength).

9.2 Characteristics of Sound

Characteristic	Depends on	Note
Pitch	**Frequency**	Higher frequency → shriller (higher) pitch; **low frequency → low (grave) pitch**
Loudness	Amplitude	Larger amplitude → louder; measured in decibel (dB)
Quality (timbre)	Waveform/overtones	Lets us distinguish two instruments at same pitch & loudness

- **Audible range:** 20 Hz – 20,000 Hz.
- **Infrasound** < 20 Hz (elephants, earthquakes); **ultrasound** > 20 kHz (bats, dolphins; used in SONAR, sonography, cleaning, flaw detection).

9.3 Echo & Reverberation

- **Echo:** reflected sound heard distinctly; requires the reflecting surface to be at least **17.2 m** away (persistence of hearing = 0.1 s; 344 × 0.1 / 2 ≈ 17.2 m).
- **Reverberation:** persistence of sound due to repeated reflections; auditoria use sound-absorbing panels to control it.

9.4 Resonance

When the forcing frequency equals the natural frequency of a body, amplitude becomes very large — resonance. Examples: shattering of a glass by a matching tone, soldiers breaking step on bridges, tuning a radio.

9.5 Doppler Effect (Class 11)

Apparent change in frequency due to relative motion of source and observer.

- Source approaching → apparent frequency **increases** (higher pitch).
- Source receding → apparent frequency **decreases**.

- Applications: radar speed guns, red-shift of galaxies, medical Doppler ultrasound.

9.6 SONAR

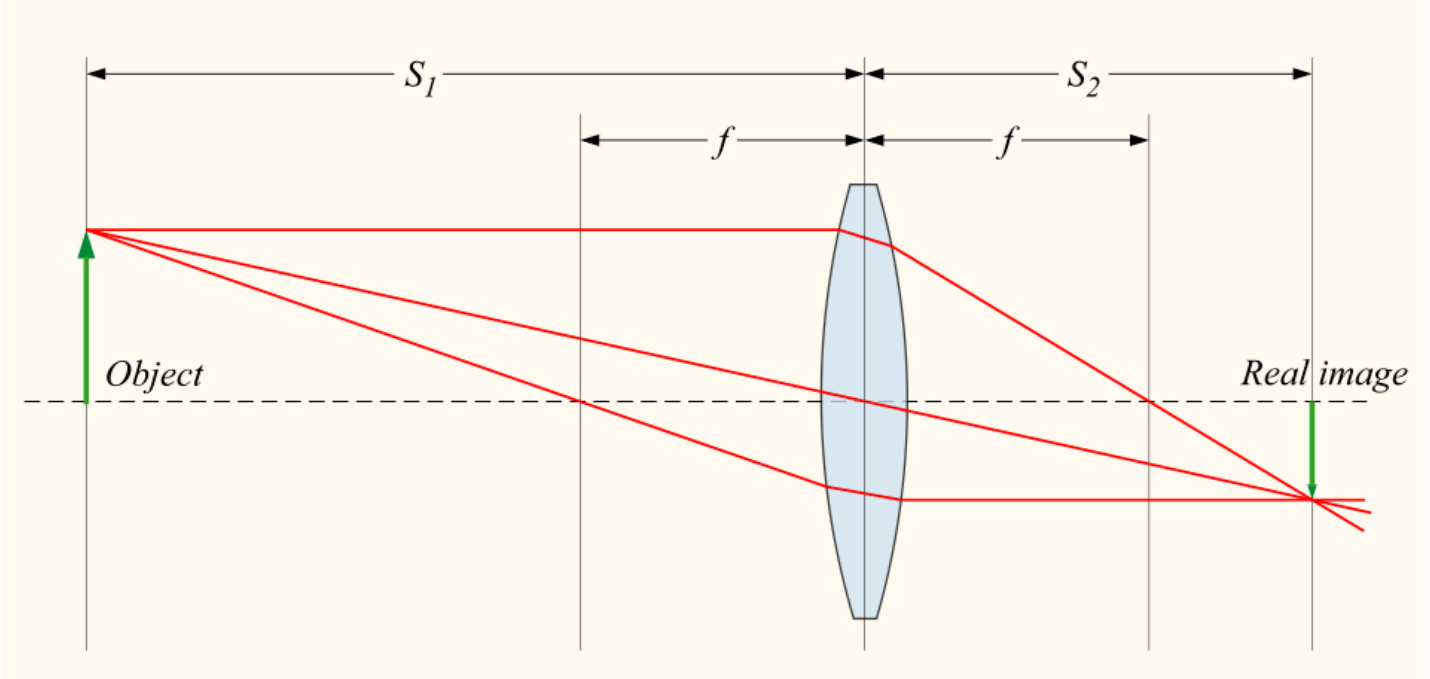
Sound **N**avigation **a**nd **R**anging — uses ultrasound to find depth/objects under water. Depth **d = v × t / 2** (t = total go-and-return time).

> **Worked Example 8.** A SONAR pulse returns from the seabed in 4 s. If the speed of sound in water is 1500 m/s, find the depth.

> d = v t / 2 = 1500 × 4 / 2 = **3000 m**.

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10. Light



Ray diagram through a convex lens.

10.1 Reflection

Laws of reflection: (i) angle of incidence = angle of reflection; (ii) incident ray, reflected ray, and normal lie in one plane. Plane mirror: image is virtual, erect, laterally inverted, same size, as far behind as the object is in front.

10.2 Spherical Mirrors

- **Concave mirror** (converging): used in torches, headlights, shaving mirrors, solar cookers, by dentists.
- **Convex mirror** (diverging): rear-view mirrors — always gives erect, diminished, virtual images with a **wide field of view**.
- **f = R/2** (focal length = half the radius of curvature).

10.3 Sign Convention (New Cartesian) — memorise this

- All distances measured **from the pole/optical centre**.
- Distances **in the direction of incident light (to the right) are positive**; against it, negative.
- Object distance **u is always negative** (object on the left).

- Heights above the principal axis positive; below, negative.
- **Concave mirror:** f negative. **Convex mirror:** f positive.
- **Convex lens:** f positive. **Concave lens:** f negative.

10.4 Mirror & Lens Formulae

	Mirror	Lens
Formula	1/v + 1/u = 1/f	1/v − 1/u = 1/f
Magnification	m = −v/u = h′/h	m = v/u = h′/h

m positive → erect & virtual image; m negative → inverted & real image; |m| > 1 enlarged, |m| < 1 diminished.

10.5 Refraction

Laws of refraction: (i) incident ray, refracted ray, normal are coplanar; (ii) **Snell's law:** sin i / sin r = n_{21} (constant for a given pair of media and colour).

- Refractive index n = c/v (speed of light in vacuum ÷ speed in medium). n_water ≈ 1.33, n_glass ≈ 1.5, n_diamond ≈ 2.42.
- Denser medium → light bends **toward the normal**; rarer medium → away from the normal.
- Effects: a pool appears shallower (apparent depth = real depth/n), a pencil in water looks bent, stars twinkle (atmospheric refraction), the Sun is visible ~2 minutes before actual sunrise and after actual sunset.

10.6 Total Internal Reflection (TIR)

Occurs when light travels from a **denser to a rarer** medium with angle of incidence **greater than the critical angle** (sin C = 1/n). Applications: **optical fibres**, sparkling of diamond, mirage, totally reflecting prisms in periscopes/binoculars.

10.7 Power of a Lens

P = 1/f (f in metres) = 100/f (f in cm), unit **dioptre (D)**.

- Convex lens → positive power; concave lens → negative power.
- Lenses in contact: P = P_1 + P_2 + …

10.8 Human Eye & Defects (quick table)

Defect	Problem	Correction
Myopia (short-sightedness)	Distant objects blurred; image forms before retina	**Concave** lens (negative power)
Hypermetropia (long-sightedness)	Near objects blurred; image behind retina	**Convex** lens (positive power)
Presbyopia	Age-related loss of accommodation	Bifocal lenses
Astigmatism	Uneven cornea curvature	Cylindrical lens

Least distance of distinct vision (near point) for a normal eye = **25 cm**.

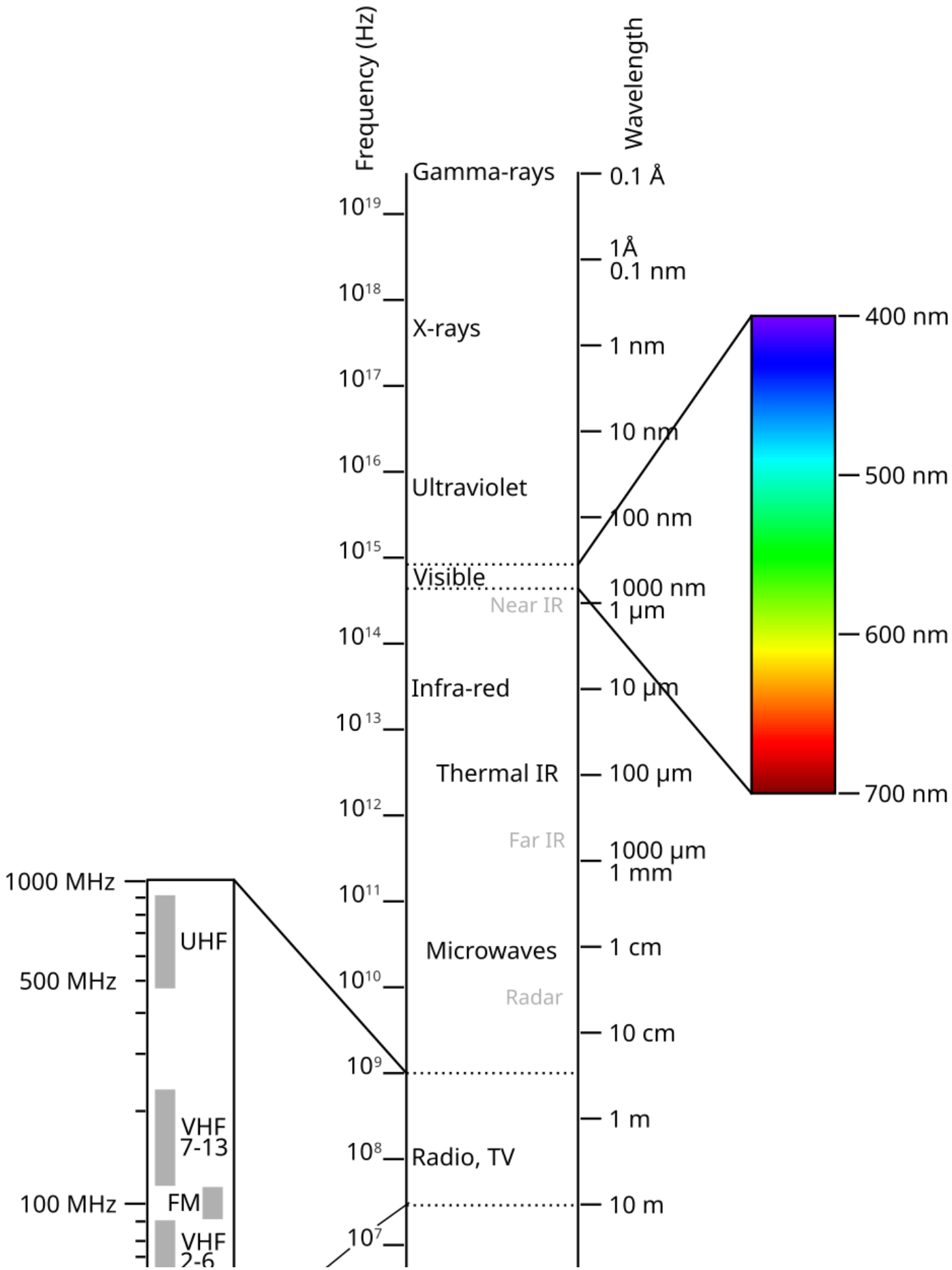
10.9 Optical Instruments (Class 12)

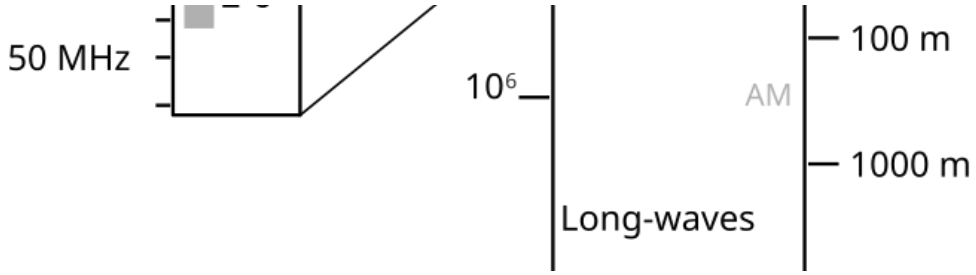
- **Simple microscope:** magnification M = 1 + D/f (image at near point D = 25 cm).
- **Compound microscope:** M = m_objective × m_eyepiece ≈ (L/f_o)(D/f_e); objective has a **short** focal length.
- **Astronomical telescope (normal adjustment): M = f_o/f_e**; objective has a **long** focal length and large aperture (light gathering & resolution).

10.10 Dispersion & Scattering

- A prism splits white light into **VIBGYOR**; **violet deviates most, red least** (n_violet > n_red). Rainbows = dispersion + TIR + refraction in raindrops.
- **Scattering ($\propto 1/\lambda^4$, Rayleigh):** blue sky (blue scatters most), red sunrise/sunset (red survives the long path), danger signals are red (least scattered). Sky appears black from the Moon/space (no atmosphere).
- **Tyndall effect:** scattering by colloidal particles makes the light path visible (sunbeam through a canopy or a dusty room).

10.11 Electromagnetic Spectrum (Class 12)


Frequency (Hz)
Wavelength
Gamma-rays
X-rays
Ultraviolet
Visible
Near IR
Infra-red
Thermal IR
Far IR
Microwaves
Radar
Radio, TV
10^{19}
10^{18}
10^{17}
10^{16}
10^{15}
10^{14}
10^{13}
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
0.1 Å
1Å
0.1 nm
1 nm
10 nm
100 nm
1000 nm
1 µm
10 µm
100 µm
1000 µm
1 mm
1 cm
10 cm
1 m
10 m
400 nm
500 nm
600 nm
700 nm
1000 MHz
500 MHz
100 MHz
UHF
VHF
7-13
FM
VHF
2-6



Electromagnetic spectrum.

In order of **increasing wavelength / decreasing frequency & energy**:

γ-rays < X-rays < Ultraviolet < Visible (400–700 nm) < Infrared < Microwaves < Radio waves

Radiation	Wavelength (approx.)	Uses
γ-rays	< 0.01 nm	Cancer therapy, sterilisation
X-rays	0.01–10 nm	Medical imaging, crystallography
Ultraviolet	10–400 nm	Water purification, forensic detection
Visible	400–700 nm	Vision
Infrared	700 nm – 1 mm	Remote controls, night vision, thermal imaging
Microwaves	1 mm – 0.1 m	Radar, microwave ovens, communication
Radio	> 0.1 m	Broadcasting, communication

UV sub-bands (know these):

Band	Wavelength	Note
UV-A	315–400 nm	Reaches Earth's surface; skin ageing
UV-B	280–315 nm	Partly absorbed by ozone; sunburn, vitamin D synthesis
UV-C	**100–280 nm**	**Shortest wavelength, highest energy — most damaging**; fully absorbed by atmosphere/ozone; used in germicidal lamps

All EM waves travel at **$c = 3 \times 10^8$ m/s** in vacuum; they are transverse and need no medium.

Worked Example 9. An object is placed 30 cm in front of a concave mirror of focal length 15 cm. Find the image position and nature.

u = −30 cm, f = −15 cm. 1/v = 1/f − 1/u = (−1/15) − (−1/30) = −1/30 ⇒ **v = −30 cm**.

m = −v/u = −(−30)/(−30) = −1. Image is **real, inverted, same size, at the centre of curvature**.

Worked Example 10. A convex lens has power +5 D. Where must an object be placed to obtain an image at 2f? Also find f.

f = 1/P = 1/5 m = **+20 cm**. A real, inverted, same-size image at 2f forms when the object is at **2f = 40 cm** from the lens (u = −40 cm).

11. Electricity

11.1 Basic Quantities (NCERT Class 10, Ch. Electricity)

- **Current I = Q/t** (ampere = coulomb/second). Conventional current flows opposite to electron flow. Charge on one electron = 1.6×10^{-19} C; 1 C ≈ 6.25×10^{18} electrons.
- **Potential difference V = W/Q** (volt = joule/coulomb), measured with a **voltmeter (in parallel)**; current with an **ammeter (in series)**.

11.2 Ohm's Law

V = IR at constant temperature (physical conditions unchanged). V–I graph for an ohmic conductor is a straight line through the origin.

Resistance of a wire: R = ρL/A — ρ = resistivity (Ω·m), a material property independent of dimensions. Resistivity of metals increases with temperature; **alloys (nichrome, constantan, manganin) have high resistivity and low temperature coefficient → used in heating elements and standard resistors**. Copper/aluminium (low ρ) → transmission wires. Tungsten (high melting point) → lamp filaments.

11.3 Series & Parallel Combinations

	Series	Parallel
Equivalent R	$R = R_1 + R_2 + R_3$	$1/R = 1/R_1 + 1/R_2 + 1/R_3$
Same for all	Current	Potential difference
Equivalent value	Larger than largest R	Smaller than smallest R
Use	Decorative series lamps	**Household wiring** (each appliance gets full mains voltage; independent switching)

Two equal resistors R in parallel → R/2; n equal resistors in parallel → R/n.

11.4 Heating Effect — Joule's Law

Heat produced: **$H = I^2Rt = VIt = (V^2/R)t$**

- $H \propto I^2$ (at fixed R): **doubling the current makes the heat four times** — the key fact behind heater elements, fuses, filament bulbs.
- **Fuse:** a low-melting-point alloy (tin–lead) wire in series that melts on overload/short circuit. Rated in amperes.
- Applications: electric iron, geyser, toaster (nichrome elements); electric bulb (tungsten filament, argon/nitrogen filling).

11.5 Electric Power

$P = VI = I^2R = V^2/R$. Energy consumed E = P × t; billed in kWh (1 kWh = 3.6×10^6 J).

> **Worked Example 11.** A 100 Ω heater element carries 2 A for 5 minutes. Find the heat produced. If the current is doubled, what happens to the heat?

> $H = I^2Rt = 2^2 \times 100 \times 300 =$ **1.2×10^5 J**.

> $H \propto I^2$, so doubling I makes the heat **4 times**: 4.8×10^5 J.

12. Magnetism & Electromagnetic Induction

12.1 Magnetic Effects of Current (NCERT Class 10)

- A current-carrying conductor produces a magnetic field around it (**Oersted's experiment**).
- Straight wire: concentric circular field lines; direction by the **right-hand thumb rule** (thumb = current, curled fingers = field).
- **Solenoid:** field like a bar magnet; uniform field inside; used with a soft-iron core to make **electromagnets**.
- Magnetic field lines are **closed curves** (N → S outside, S → N inside a magnet) and **never intersect**. SI unit of magnetic field: tesla (T).

12.2 Force on a Current-Carrying Conductor

$F = BIL \sin\theta$ (maximum when the conductor is perpendicular to the field).

Fleming's left-hand rule (motor rule): **F**ore-finger = **F**ield, **C**entre finger = **C**urrent, Thu**M**b = **M**otion (force).

12.3 Electromagnetic Induction

- **Faraday's laws:** an EMF is induced whenever the magnetic flux linked with a circuit changes; induced EMF ∝ **rate of change of flux** ($\varepsilon = -d\Phi/dt$).
- **Lenz's law:** the induced current opposes the change that produces it (consequence of energy conservation) — the minus sign in Faraday's law.
- **Fleming's right-hand rule** (generator rule) gives the direction of induced current.

12.4 Motor, Generator, Transformer

Device	Converts	Principle	Key part
Electric motor	Electrical → mechanical	Force on a current-carrying coil in a magnetic field	Split-ring commutator (DC) reverses current each half turn
Generator (dynamo)	Mechanical → electrical	Electromagnetic induction	AC: slip rings; DC: split-ring commutator
Transformer	AC voltage level up/down	Mutual induction (**AC only**)	$V_s/V_p = N_s/N_p$; step-up: $N_s > N_p$

- India's domestic supply: **220 V, 50 Hz AC** (direction reverses every 1/100 s → 50 cycles/s).
- Wiring: **live (red/brown), neutral (black/light blue), earth (green)**. The earth wire protects users from shocks via metallic appliance bodies. **MCBs/fuses** are placed in the live wire.
- AC is preferred for transmission because voltage can be stepped up (low current → low I^2R line loss).

13. Modern Physics

13.1 Atomic Models

Model	Key idea	Shortcoming
Thomson (1898)	"Plum-pudding": electrons embedded in a positive sphere	Could not explain scattering results
Rutherford (1911)	α-scattering → tiny, dense, positive **nucleus**; electrons revolve around it	Orbiting electron should spiral in (instability); no line spectra
Bohr (1913)	Electrons in fixed **stationary orbits** with quantised angular momentum ($mvr = nh/2\pi$); energy emitted/absorbed only on jumps ($E_2 - E_1 = h\nu$)	Works well only for hydrogen-like atoms

Nucleus discovered by Rutherford; **proton** by Rutherford (1919); **neutron by Chadwick (1932)**; electron by J. J. Thomson (1897).

13.2 Radioactivity

Spontaneous emission of radiation by unstable nuclei (discovered by **Henri Becquerel**, 1896; term coined by Marie Curie).

Radiation	Nature	Charge	Penetrating power	Ionising power	Stopped by
α (alpha)	Helium nucleus ($_2He^4$)	+2	Lowest	Highest	Sheet of paper
β (beta)	Fast electrons	−1	Medium	Medium	Few mm of aluminium
γ (gamma)	EM waves (photons)	0	**Highest**	Lowest	Thick lead/concrete

- α-decay: mass number ↓ 4, atomic number ↓ 2. β-decay: atomic number ↑ 1, mass number unchanged. γ-emission: no change in A or Z.
- SI unit of activity: **becquerel (Bq)**; older unit: curie (Ci).

13.3 Half-Life

Time in which half the radioactive nuclei decay. After n half-lives, remaining fraction = **$(1/2)^n$**; $N = N_0(1/2)^{(t/T½)}$.

> **Worked Example 12.** A radioactive sample of 80 g has a half-life of 5 days. How much remains after 20 days?

> Number of half-lives $n = 20/5 = 4$. Remaining $= 80 \times (1/2)^4 = 80/16 =$ **5 g**.

13.4 Fission vs Fusion

	Nuclear fission	Nuclear fusion
Process	Heavy nucleus (^{235}U) splits into lighter nuclei on neutron capture	Light nuclei (hydrogen isotopes) combine into helium
Conditions	Room conditions possible (chain reaction, moderated)	Extremely high temperature & pressure (thermonuclear)
Energy per unit mass	Large	**Larger** than fission
Where used/found	Nuclear reactors, atom bomb	**The Sun and stars**, hydrogen bomb
Mass–energy	Both governed by **$E = mc^2$** (mass defect converts to energy)	

Reactor essentials: fuel (enriched U), **moderator** (heavy water/graphite — slows neutrons), **control rods** (cadmium/boron — absorb neutrons), coolant.

13.5 X-rays

Discovered by **W. C. Röntgen (1895)**. Produced when fast electrons strike a heavy metal target (tungsten). High-frequency EM waves; penetrate flesh but not bone/metal — medical imaging, security scanning, crystal structure analysis.

13.6 Photoelectric Effect (Class 12 basics)

Emission of electrons from a metal surface when light of frequency above a **threshold frequency ν_0** falls on it (explained by **Einstein**, Nobel Prize 1921).

- **Einstein's equation:** $h\nu = \varphi + KE_{max}$ (φ = work function = $h\nu_0$).
- Number of photoelectrons ∝ **intensity**; maximum KE ∝ **frequency** (not intensity).
- Instantaneous emission; demonstrates the **particle (photon) nature of light**. Photon energy $E = h\nu = hc/\lambda$, $h = 6.63 \times 10^{-34}$ J·s.
- Applications: photocells, automatic doors, solar panels, light meters.

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14. PYQ Corner — RRB JE CBT-2 2025 (Physics Questions Solved)

PYQ 1. A concave lens has focal length 2 cm. Its power is:

Concave lens ⇒ f is negative: f = −2 cm.

P = 100/f(cm) = 100/(−2) = **−50 D**. ✓ Negative power confirms a diverging (concave) lens.

PYQ 2. The property of a metal by which it can be drawn into thin wires is:

Ductility. (Malleability = beaten into sheets; gold and silver are the most ductile/malleable metals.)

PYQ 3. A body starts from rest and accelerates uniformly; using $v^2 = u^2 + 2as$, find its kinetic energy after covering distance s under acceleration a (mass m).

$v^2 = 0 + 2as \Rightarrow KE = \frac{1}{2}mv^2 = \frac{1}{2}m(2as) =$ **mas** — numerically equal to the work done, W = Fs = (ma)s. This is exactly the KE derivation in §4.2; CBT-2 gives numbers (e.g., m = 2 kg, a = 5 m/s², s = 10 m ⇒ KE = 2 × 5 × 10 = 100 J).

PYQ 4. Low frequency of a sound wave corresponds to:

Low pitch (grave/dull sound). Pitch is the perceptual attribute determined by frequency — high frequency = shrill/high pitch, low frequency = low pitch (§9.2).

PYQ 5. An object is placed 15 cm from a convex lens of focal length 25 cm. Find the image distance and nature.

u = −15 cm, f = +25 cm (object inside the focus, since 15 < 25).

1/v = 1/f + 1/u = 1/25 + (−1/15) = (3 − 5)/75 = −2/75 ⇒ **v = −37.5 cm**.

Negative v ⇒ image on the **same side as the object**: **virtual, erect, magnified** (m = v/u = −37.5/−15 = +2.5). A convex lens acts as a magnifying glass when the object is within its focal length.

PYQ 6. In an electric heating element, the heat produced is proportional to:

The square of the current (H ∝ I^2) — Joule's law H = I^2Rt (§11.4). Doubling I quadruples the heat.

PYQ 7. Which UV band is the most damaging?

UV-C (100–280 nm) — shortest wavelength ⇒ highest frequency and photon energy ⇒ most damaging. It is fortunately absorbed completely by the ozone layer/atmosphere (§10.11).

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15. Practice MCQs (25)

Q1. The SI unit of luminous intensity is:

(a) lumen (b) lux (c) candela (d) weber

Q2. The area under a velocity–time graph gives:

(a) acceleration (b) displacement (c) force (d) momentum

Q3. A body moving in a circle at constant speed has:

(a) constant velocity (b) zero acceleration (c) acceleration toward the centre (d) acceleration along the tangent

Q4. For maximum horizontal range, a projectile should be launched at:

(a) 30° (b) 45° (c) 60° (d) 90°

Q5. Inertia of a body is measured by its:

(a) weight (b) velocity (c) mass (d) volume

Q6. Rocket propulsion is based on conservation of:

(a) energy (b) mass (c) linear momentum (d) angular momentum

Q7. Which type of friction is the smallest?

(a) static (b) sliding (c) rolling (d) limiting

Q8. A body of mass 4 kg moves at 3 m/s. Its kinetic energy is:

(a) 6 J (b) 12 J (c) 18 J (d) 36 J

Q9. 1 kWh equals:

(a) 3.6×10^5 J (b) 3.6×10^6 J (c) 3.6×10^7 J (d) 746 J

Q10. The value of g is maximum at:

(a) the equator (b) the poles (c) the centre of the Earth (d) a mountain top

Q11. Kepler's second law (equal areas in equal times) is a consequence of conservation of:

(a) energy (b) linear momentum (c) angular momentum (d) mass

Q12. Hydraulic brakes work on:

(a) Archimedes' principle (b) Pascal's law (c) Bernoulli's theorem (d) Hooke's law

Q13. A hydrometer is used to measure:

(a) humidity (b) pressure (c) density of liquids (d) depth of the sea

Q14. Which is more elastic?

(a) rubber (b) steel (c) putty (d) clay

Q15. Water has maximum density at:

(a) 0 °C (b) 4 °C (c) 100 °C (d) −4 °C

Q16. Steam at 100 °C causes more severe burns than boiling water at 100 °C because steam has additional:

(a) temperature (b) specific heat (c) latent heat of vaporisation (d) pressure

Q17. The minimum distance for hearing a distinct echo (v ≈ 344 m/s) is about:

(a) 10 m (b) 17.2 m (c) 34.4 m (d) 344 m

Q18. When a sound source approaches a stationary listener, the apparent frequency:

(a) decreases (b) increases (c) is unchanged (d) becomes zero

Q19. Total internal reflection can occur when light travels from:

(a) air to glass (b) air to water (c) water to air (d) vacuum to glass

Q20. A person cannot see distant objects clearly. The defect and its correction are:

(a) myopia — convex lens (b) myopia — concave lens (c) hypermetropia — concave lens (d) hypermetropia — convex lens

Q21. In an astronomical telescope (normal adjustment), magnification equals:

(a) f_e/f_o (b) f_o/f_e (c) $f_o \times f_e$ (d) $f_o + f_e$

Q22. Which colour of light deviates most through a glass prism?

(a) red (b) yellow (c) green (d) violet

Q23. Three 6 Ω resistors are connected in parallel. The equivalent resistance is:

(a) 18 Ω (b) 6 Ω (c) 3 Ω (d) 2 Ω

Q24. A transformer works on:

(a) DC only (b) AC only (c) both AC and DC (d) electrostatic induction

Q25. The half-life of a radioactive element is 10 days. The fraction remaining after 30 days is:

(a) 1/3 (b) 1/4 (c) 1/8 (d) 1/16

Answer Key with One-Line Reasons

Q	Ans	Reason
1	c	Candela is the SI base unit of luminous intensity
2	b	Area of v–t graph = displacement
3	c	Centripetal acceleration a = v^2/r toward the centre
4	b	R = u^2 sin 2θ/g is maximum when 2θ = 90°
5	c	Mass is the measure of inertia
6	c	Momentum of gases backward = momentum of rocket forward
7	c	Rolling < sliding < static friction
8	c	KE = ½ × 4 × 3^2 = 18 J
9	b	1 kWh = 1000 W × 3600 s = 3.6 × 10^6 J
10	b	Earth is flattened at the poles (smaller R → larger g)
11	c	Equal-area law ⇔ constant angular momentum
12	b	Pressure transmitted undiminished in an enclosed fluid
13	c	Hydrometer floats per liquid density (floatation)
14	b	Steel needs more stress for the same strain
15	b	Anomalous expansion — densest at 4 °C
16	c	Extra 540 cal/g released on condensation
17	b	d = v × 0.1/2 ≈ 17.2 m
18	b	Doppler effect — approach raises frequency
19	c	TIR needs denser → rarer with i > critical angle
20	b	Myopia corrected by a diverging (concave) lens
21	b	M = f_o/f_e
22	d	n_violet is the largest; violet bends most
23	d	R/n = 6/3 = 2 Ω
24	b	Mutual induction needs changing flux — AC only
25	c	n = 3 half-lives ⇒ $(1/2)^3$ = 1/8

16. Rapid-Revision Formula Sheet

Mechanics

- $v = u + at$; $s = ut + \frac{1}{2}at^2$; **$v^2 = u^2 + 2as$**; $s_n = u + a(2n-1)/2$
- $F = ma$; $p = mv$; Impulse $= F\Delta t = \Delta p$
- $W = Fs\cos\theta$; **$KE = \frac{1}{2}mv^2 = p^2/2m$**; $PE = mgh$; $P = W/t = Fv$
- $F = Gm_1m_2/r^2$; $g = GM/R^2$; $W = mg$; $v_{escape} = \sqrt{(2gR)} = 11.2$ km/s; $T^2 \propto r^3$
- $P = F/A$; $P = h\rho g$; Buoyant force = weight of fluid displaced
- Stress/strain = modulus; $F = kx$; spring $PE = \frac{1}{2}kx^2$

Heat & Sound

- $Q = mc\Delta T$; $Q = mL$; $C/5 = (F-32)/9 = (K-273)/5$
- $v = f\lambda$; echo distance ≥ 17.2 m; SONAR depth $d = vt/2$
- Audible range 20 Hz–20 kHz; pitch ↔ frequency; loudness ↔ amplitude

Light

- Mirror: $1/v + 1/u = 1/f$, $m = -v/u$, $f = R/2$
- Lens: $1/v - 1/u = 1/f$, $m = v/u$
- **$P = 1/f(m) = 100/f(cm)$** dioptre; lenses in contact $P = \Sigma P_i$
- $n = c/v = \sin i/\sin r$; $\sin C = 1/n$
- Telescope $M = f_o/f_e$; simple microscope $M = 1 + D/f$
- EM spectrum (λ increasing): γ < X < UV < visible < IR < microwave < radio; UV-C most energetic UV
- $c = 3 \times 10^8$ m/s

Electricity & Magnetism

- $I = Q/t$; $V = W/Q$; **$V = IR$**; $R = \rho L/A$
- Series $R = \Sigma R_i$; parallel $1/R = \Sigma 1/R_i$
- **$H = I^2Rt = VIt = V^2t/R$**; $P = VI = I^2R = V^2/R$; 1 kWh $= 3.6 \times 10^6$ J
- $F = BIL\sin\theta$; Fleming's left hand → motor; right hand → generator
- $\varepsilon = -d\Phi/dt$ (Faraday + Lenz); transformer $V_s/V_p = N_s/N_p$

Modern Physics

- Bohr: $mvr = nh/2\pi$; $\Delta E = h\nu$
- $N = N_0(\frac{1}{2})^{(t/T\frac{1}{2})}$; after n half-lives, fraction left $= (\frac{1}{2})^n$
- $E = mc^2$; photon $E = h\nu = hc/\lambda$; Einstein: $h\nu = \varphi + KE_{max}$
- Penetration: γ > β > α; ionisation: α > β > γ

End of GA-01. Next: GA-02 Chemistry (NCERT).